



Station 1: Wave Erosion



How do waves change a coastline?



Procedure:

1. Build a "beach" - pack sand to make a gentle slope in one end of tray
2. Pour water carefully into the other end (about 2 cm deep)
3. Use your hand to create 10 gentle waves toward the beach
4. Observe: What happens to the sand?
5. Create 10 MORE waves (stronger this time)
6. Measure how far the beach moved back



Record in Your Journal:

- Draw "before" and "after" sketches
- How many cm did the beach erode?
- Where did the sand go?
- What would happen with BIGGER waves?



Real World: This is exactly what happens on ocean coastlines during storms! Over time, waves reshape beaches and cliffs.